

Section A : Data fundamentals

(1) Database : A database is an organized collection of structured information stored electronically. It is essential because it allows organizations to safely store, retrieve, and manage massive amount of information efficiently.

Examples :

- Banks use databases to store customer accounts and transactions
- Online stores use databases to store products, customers and orders.

(2) Data Warehouse : A data warehouse is a central repository designed specifically to store historical data pulled from multiple sources. While a regular database handles real time daily transactions, a warehouse is optimized for running complex analytics. Companies use it to generate business intelligence reports without slowing down their live operational systems.

(3) Data Lake : A data lake is a vast storage system that holds raw, unformatted data in its native state until it is needed. Unlike a data warehouse that strictly stores cleaned and structured data, a data lake accepts everything. A business prefers a data lake when storing massive volumes of unstructured data, like social media feeds, for future machine learning projects.

(4) Data Mart : A data mart is a smaller, specialized subset of a data warehouse. It is focused entirely on a specific business line or department, such as Sales, or finance and is typically used by department managers to quickly access relevant data without querying the entire warehouse.

(5) Data Types : Structured data is highly organized into tabular formats, like an Excel spreadsheet of customer names. Semi-structured data contains organizational tags but lacks a rigid table format, such as a JSON file used in web development. Unstructured data has no predefined format, like an MP3 audio recording or a JPEG image.

(6) Data Modeling : Data Modeling is the process of creating a visual blueprint of how data is collected, stored, and ~~collected~~ connected within a system. It is critical because it ensures the database is built logically and can scale effectively.

Examples :

- Conceptual model - gives a high level view of the data and its relationships.
- Physical model - shows how the data will actually be stored in the database.

(7) Data Mining : Data Mining is a practice of analyzing large datasets to uncover hidden patterns, trends and anomalies. Businesses use it to predict consumer behaviour and optimize strategies.

Examples :

- Banks use data mining to detect fraudulent transactions.
- Retailers use it to identify products that customers often buy together.

(8) DBMS: A Database Management System is a software used to manage and interact with data

DBMS	Brief description	Example
MySQL	Relation database system	Websites
PostgreSQL	Open-source relational database	Business application
Oracle	Enterprise database system	Banking
MongoDB	Document-based NoSQL database	Modern web application

(9) ETL Pipeline: ETL stands for Extract, Transport and Load.

- Extract: Collect data from different sources
- Transform: Clean, organize, and change the data into a useful format
- Load: Store the prepared data in a target system such as a data warehouse.

(10) Storage formats:

- CSV is a simple text format used for lightweight data sharing and basic spreadsheets.
- JSON ~~used~~ uses key-value pairs and is the standard format for web APIs.
- Parquet is a columnar format optimized for running fast analytics on huge datasets.
- Avro is a row-based format ideal for streaming data where the structure might change over time.

Section 8: Artificial Intelligence & Machine Learning

(11) Artificial Intelligence: Artificial Intelligence is the simulation of human intelligence processes by computer systems, enabling them to learn, reason and solve problems.

- Narrow AI (Weak AI) is designed to perform on specific task, such as a chess playing computer program
- General AI (Strong AI) is a theoretical concept where a machine possesses human-like cognitive abilities and can independently learn to perform any intellectual task a human can.

(12) Machine Learning: Machine Learning (ML) is a branch of AI that focuses on building system that can learn from and improve upon data without being explicitly programmed. In traditional programming, a human writes the rules to process data into answers, in ML the machines takes data and answers to figure out the rules.

- Supervised Learning trains on labeled where the correct answers are provided (e.g. sorting emails into spam or inbox).
- Unsupervised Learning finds hidden patterns in unstructured, unlabeled data (e.g. grouping shoppers by purchasing habits.)
- Reinforcement Learning trains models through a system of trial and error, using rewards and penalties to achieve a goal (e.g. teaching a computer to play a video game)

(13) Deep Learning and Neural Networks: Deep learning is a specialized subset of machine learning based on artificial neural networks, which are algorithms inspired by the structure of the human brain. Neural networks consist of multiple layers of interconnected nodes (neurons). They work by receiving data in the input layer, passing it through "hidden layers" that mathematically adjust weights and biases to extract complex features and finally delivering a prediction through the output layer.

(14) Natural Language Processing (NLP): Natural Language Processing is a field of AI that gives computers the ability to understand, interpret and generate human language in a meaningful way. Examples include:

- Voice assistants (like Siri or Alexa) understanding spoken commands
- Email platforms automatically summarizing long messages
- Translation apps (like Google Translate) instantly converting text between languages.

(15) Large Language Models (LLMs): An LLM is a massive deep learning model that has been trained on vast amounts of text data to understand and generate human-like language. It differs from standard ML models because of its sheer scale and its ability to handle general, open-ended conversational tasks rather than one specific function. Examples:

- OpenAI's ChatGPT used for drafting text, coding and brainstorming
- Google's Gemini used for complex reasoning, research and multimodal analysis.

(16) Generative AI: Generative AI refers to algorithms capable of creating new, original content such as text, image or audio by learning patterns from existing training data. At a high level, it works by predicting the next logical sequence in a pattern based on a user's prompt like guessing the next word in a sentence or the next pixel in an image. Three tools include:

- Midjourney (Generates high-quality images from text descriptions)
- GitHub Copilot (Generates programming code to assist developers)
- Suno (Generates original music tracks and lyrics).

(17) Training Datasets & Quality: A training dataset is the initial collection of data used to teach a machine learning model how to recognize patterns and make decisions. The quality of this data is critical because a model's output is entirely dependent on what it learns, often referred to as "garbage in, garbage out". If the training data is biased or incorrect, the AI model will learn and amplify those flaws, resulting in skewed, unfair or highly inaccurate predictions in the real world.

(18) Computer Vision: Computer Vision is a field of AI that allows machine to extract meaningful information from digital images, video, and visual inputs. Machines "see" an image by breaking it down into a massive grid of pixels, reading the numerical color values of each pixel, and using neural networks to identify patterns like lines, edges and shapes. Two real-world uses:

- Healthcare: Analyzing X-rays & other medical images
- Self-driving cars: Detecting road, traffic signs, pedestrians, and other vehicles.

(19) API: An API is a set of rules that allows different software applications to communicate with each other. A real world analogy is a waiter in a restaurant: you give the waiter your order, the waiter takes it to the kitchen and then bring you your food. APIs is important because they allow developers to easily integrate external features, like adding Google Maps to a delivery app, without building the systems from scratch.

(20) API keys: An API key is a unique code passed in by an application to authenticate its identity when calling an API. Its purpose is to control on track who is using the service. It should be stored securely in hidden environment variable (like a .env file) and never hardcoded directly into the code. If an API key is exposed publicly, malicious users can steal data, exhaust usage limit, and run up massive billing charges on the owner's account.

(21) REST API: REST stands for Representational State Transfer, which is a set of architectural rules for designing networked applications. It relies on standard HTTP methods to interact with data. GET is used to retrieve or read data from a server. POST is used to send new data to the server to create a record. PUT is used to update or modify an existing record. DELETE is used to remove a record from the server.

(22) JSON: ~~JSON~~ JSON is a lightweight text format used for storing and transporting data. It is commonly used with APIs because it is easy for humans to read and write and simple for machine to parse and generate across almost all programming language.

A simple example is :

```
{ "name" : "Sanele",  
  "age" : 24,  
  "role" : "Data Analyst" }
```

(23) Prompt Engineering: Prompt engineering is the practice of designing clear and highly specific input (prompts) to guide Generative AI models into producing optimal output. It is an important skill because AI models lack context, instructions reduce errors and hallucinations.

• Example 1: A poor prompt is "Write an email". An improved version is "Write a concise, professional email to my manager requesting approval for next week's annual leave."

• Example 2: A poor prompt is "Explain databases." An improved version is "Explain the difference between a data warehouse and a data lake using a simple analogy suitable for a beginner."

(24) MAP Function: A MAP function is a programming command that takes a specific mathematical operation or rule and applies it sequentially to every single item in a list or array, returning a new list with the modified items.

Further example:

```
numbers = [1, 2, 3, 4]
```

```
result = list(map(lambda x: x*2, numbers))
```

The result is

[2, 4, 6, 8]

A map function multiplies each number by 2.

(25) Cloud Computing: Cloud computing is the delivery of computing services, such as servers, storage, and databases over the internet instead of relying on local hard drives. The three main service models are:

Model	Meaning	Example
• IaaS	Infrastructure as a Service	Amazon EC2
• PaaS	Platform as a Service	Google App Engine
• SaaS	Software as a Service	Microsoft 365

IaaS provides infrastructure, PaaS provides a platform for developing applications and SaaS provides ready-to-use software.

(26) Version Control & Git: A Version Control System is a software tool that tracks and manages changes to code over time, allowing teams to collaborate and revert to previous versions if mistakes happen. Git is the most widely used distributed version control system. A commit is a saved snapshot of your code at a specific point in time. A branch is a parallel, independent version of the main project where a

developer can safely test new features without breaking the core code

(27) Jupyter Notebook: A Jupyter Notebook is an open-source web application that allows users to create and share documents containing live code, mathematical equations, visualizations and narrative text. It is most commonly used with the Python programming language. It is incredibly popular among Data Analysts because it allows to execute code block-by-block and see charts or data table immediately below the code.

(28) Python: Python is a high-level programming language known for its simple, readable syntax. It is the most popular language for Data Analytics and AI because it is easy to learn and possesses a massive ecosystem of specialized tools built specifically for data processing. Four common Python libraries are:

- Pandas - used for data manipulation and analysis
- NumPy - used for numerical and mathematical operations
- Matplotlib - used to create graphs and visualizations
- Scikit-learn - used for machine learning

Section D: Data Ethics, Governance & Careers

(20) Data Privacy & GDPR: Data privacy is the practice of handling personal information responsibly and securely, ensuring individuals maintain control over their own digital footprint. It is crucial because it protects people from identity theft, targeted fraud, and invasive surveillance. The GDPR is a strict data privacy law drafted by the European Union, it applies to any global organization that targets or collects data

related to citizens within the EU, regardless of where the company is physically located.

(30) Data Ethics: Data ethics involve the moral principles guiding how data is collected, stored and utilized. Three problematic examples include:

- A technology company secretly selling user location data to advertisers, which fundamentally violates consumer trust and privacy
- Using facial recognition software trained on non-diverse ~~dataset~~ datasets, resulting in disproportionate false identifications for minorities
- An algorithm automatically rejecting loan application based on postal codes, which perpetuates historical redlining and systemic financial discrimination.

(31) Data Governance: Data governance is an organizational framework that establishes the policies, procedures and responsibilities for managing data assets securely and accurately across a company. A strong framework includes clear data quality standards, strict security ~~for~~ protocols to prevent unauthorized access, defined roles (like data stewards) who maintain data integrity, and compliance mechanisms to ensure adherence to legal regulations.

(32) AI Bias: AI bias occurs when a machine learning algorithm produces prejudiced or systematically unfair outcomes ~~for~~ because of erroneous assumptions embedded in its training process. A real-world example is an AI hiring tool trained on historical resumes submitted predominantly by men

- the model might incorrectly learn to penalize resumes

containing the word "women's" (like "women's chess club").

The consequences include perpetuating workplace inequality and exposing organizations to severe legal and reputational damage.

(33) Data Careers Comparison :

Role	Main responsibility	Common skills/tools
• Data Analyst	Analyzes data and creates reports	Excel, SQL, Power BI
• Data Scientist	Builds model and find patterns	Python, R, Machine Learning
• Data Engineer	Builds and maintains data systems and pipelines	SQL, Python, Cloud tools
• AI Engineer	Develops and deploys AI solutions	Python, ML, Deep Learning

(34) Business Intelligence (BI) : Business Intelligence refers to the strategies and technologies used by enterprises to analyze current and historical business data to drive actionable insights. Two popular BI tools are Power BI and Tableau. They help businesses make data-driven decisions by transforming raw spreadsheets into interactive, visual dashboards, allowing decision makers to easily spot market trends, track performance metrics and identify operational bottlenecks at a glance.

Section E : Personal Reflection

(35) Most Interesting Concept : The concept I found most interesting was Machine Learning. I found it interesting because computers can learn from data and use them to make predictions without being explicitly programmed for every situation. It showed me how mathematics, statistics and programming can be combined to solve real-world problems.

(36) Most Difficult Concept : The concept I found most difficult was Data Governance because it involves many areas such as data quality, privacy, security, policies and responsibilities. I used online resources from IBM and other technology websites to understand how organisations manage and protect their data.

(37) Impact on future Career : Data and AI will be central to optimising operational workflows, especially ~~within~~ within public sector analytics. Commencing work within a provincial governance environment, like the Gauteng Office of the Premier, requires leveraging accurate data to influence evidence-based policy. Applying structured machine learning models and robust data pipelines to regional ~~to datasets~~ datasets will directly improve operational efficiency and help track key service delivery metrics effectively.